

Reverse Engineering Turntable



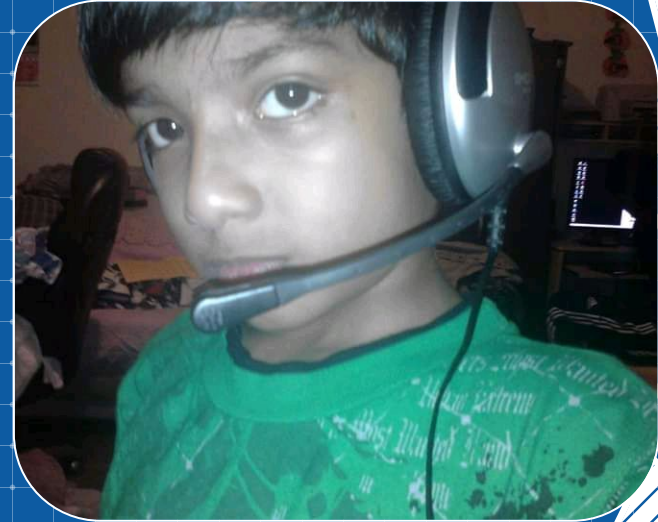
Turntable

ME 14500 | Fall 2025 | Md Rahman, Fairuz Shrabon | Group G

Group G



Md Rahman



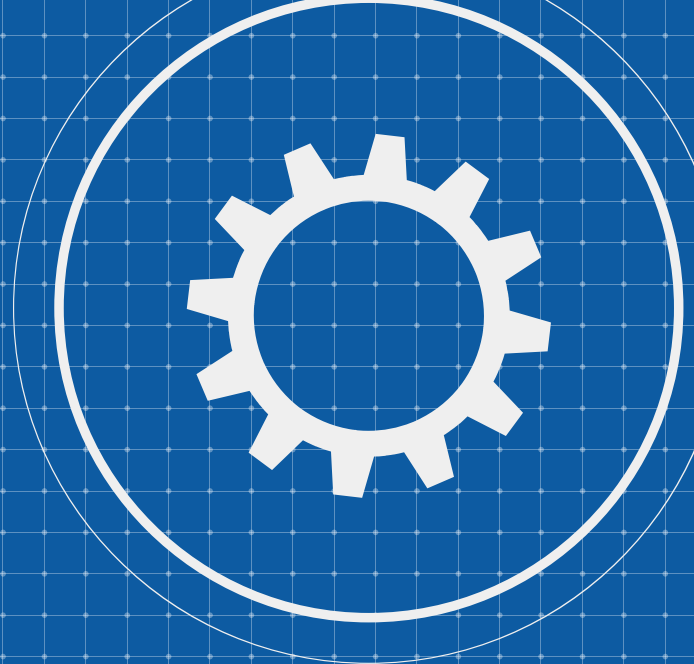
Fairuz Shrabon



01

Introduction

Introduction – Rotating Display Stand





Toy Introduction – Rotating Display Stand

Purpose

- Motorized 360° stand for cups and small products
- Provides smooth rotation for photos and videos

Age group / users

- Teens and adults, not young kids
- DIY epoxy tumbler makers and small online sellers



About \$10.99 on Amazon

- First released around **2021** as a budget 360° turntable for product photography
- \$5.27 on Alibaba, typically drop shipped. (108% markup)



Cost Analysis

Cost breakdown (factory level)

- **ABS + PC plastic material:** ~\$0.50–0.75
- **Injection molding + tooling (amortized):** ~\$0.35–1.10
- **Motor + electronics + hardware:**
 - High torque DC gear motor: ~\$1.10–2.80
 - Controller PCB + switch + speed selector: ~\$0.20–0.60
 - 12V power adapter + DC jack: ~\$0.80–1.80
 - Screws, rubber feet, foam pads: ~\$0.05–0.15

Assembly, QA, packaging, overhead: ~\$0.60–1.60

- **Typical factory gross margin:** 20–40%

Implied COGS: **\$4.08 to \$2.20**

- Gross profit: **\$0.39 to \$1.44**

(Depending on # of units sold)

Specifics:

- **Material:** ABS + PC blend, injection molded housing and top plate
- **Input power:** **DC 12V** (external adapter, not USB)
- **Load capacity:** 10–15 kg, requires higher torque motor and gearbox
- **Unit mass:** ≈ 0.35 kg



Image from Alibaba



Video of Toy in Operation



The turntable can spin clockwise or counter-clockwise.

It comes with a friction pad on top so items don't slip.

It also has a slot to plug it in with USB-C if you don't want to use batteries.



Reverse Engineering – What We Did

What is reverse engineering?

- Start from a finished product and work **backwards** to figure out how it is built
- Study **geometry, materials, and mechanisms** inside
- Use that understanding to **document** the design and **improve** it

Step 1 – Select & study: pick a low cost toy/tool, record how it moves and behaves

Step 2 – Disassemble & measure: open it, measure all key parts and create 2D hand sketches

Step 3 – CAD modeling: rebuild the toy in **SolidWorks** with 2D drawings and 3D models

Step 4 – Improve & test: redesign weak parts, 3D print them, and test them on the original toy





External Views



Top view – rotating plate and remote



Front view – CW / OFF / CCW switch and DC 5V port



Bottom view – battery compartment and drive shaft



Internal Views After Disassembly



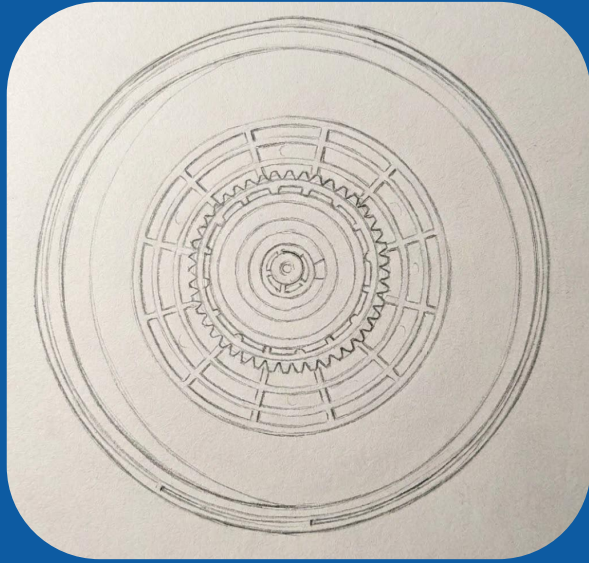
Base – motor, gear train,
PCB, wiring



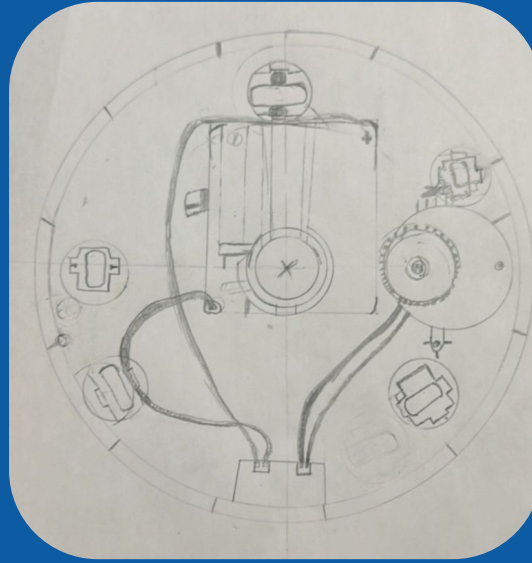
Top plate – driven gear and
shaft (blue lubricant)



2D Technical Drawing



Top projection – rotating plate and gear ring

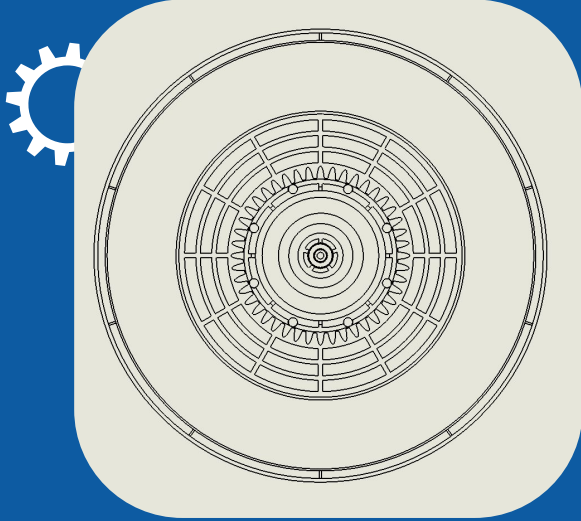


Bottom projection – interior mechanism layout

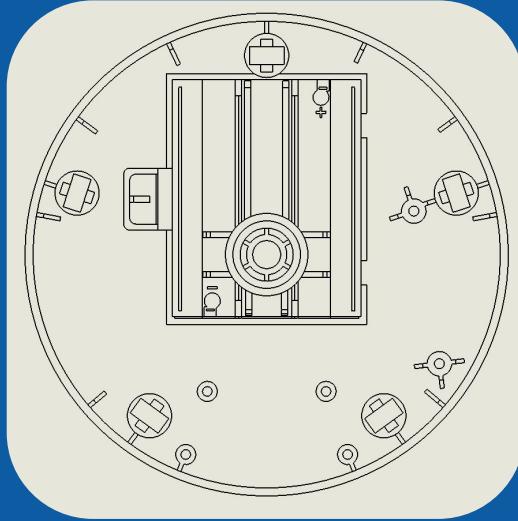
- Drawn from **real measurements** after disassembly
- Uses **circles, arcs, and lines** to simplify complex geometry
- Shows **overall size** and **relative position** of motor, gears, and wiring



2D SolidWorks



Top projection – rotating plate and gear ring



Bottom projection – interior mechanism

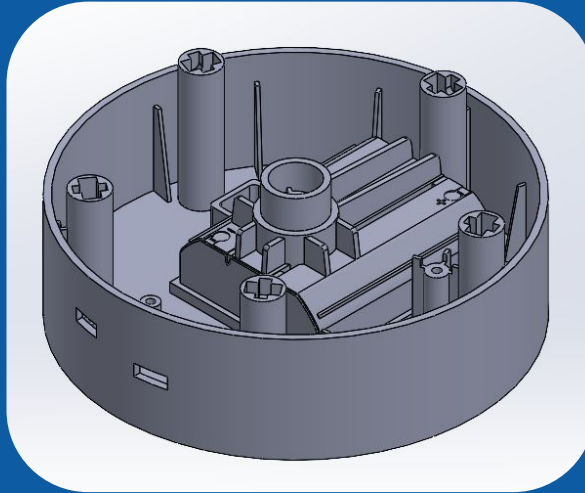
- Recreated our hand sketches in SolidWorks
- Used parametric dimensions to control key sizes
- Basis for 3D models and section views later



3D SolidWorks



Rotating top platform with
gear ring

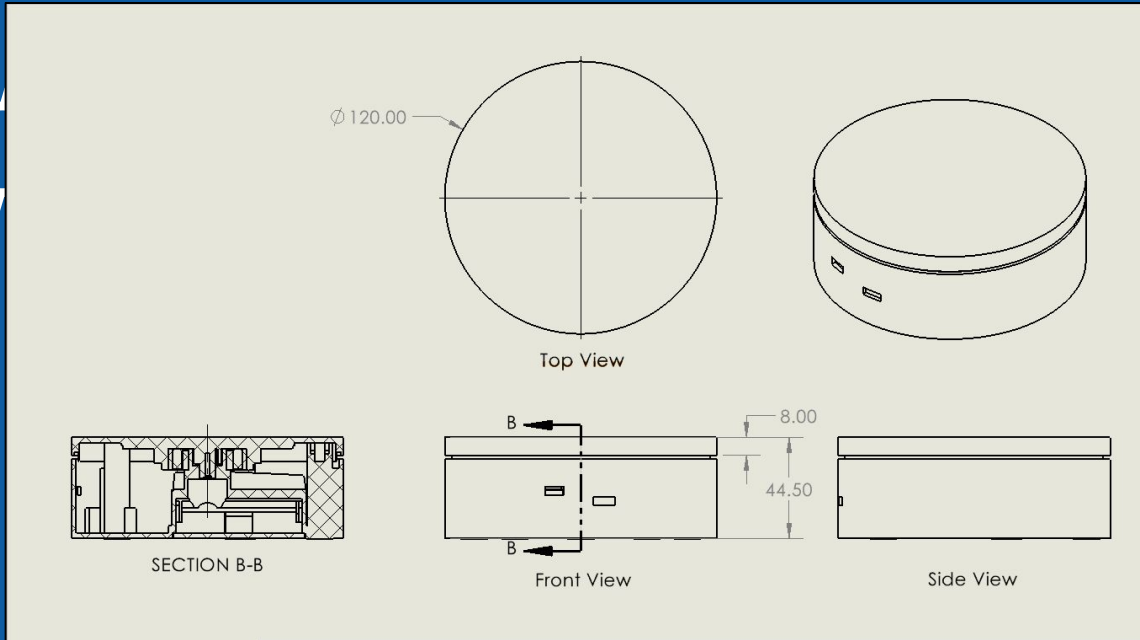


Base housing with motor
mount and internal supports

- Converted our measured sketches into full 3D parts
- Modeled the top platform and the base housing as separate components
- Used these models later for section views, clearance checks, and redesigns for 3D printing



Completed Drawing

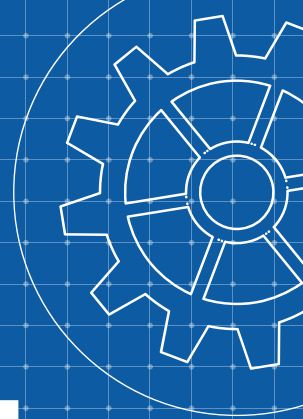
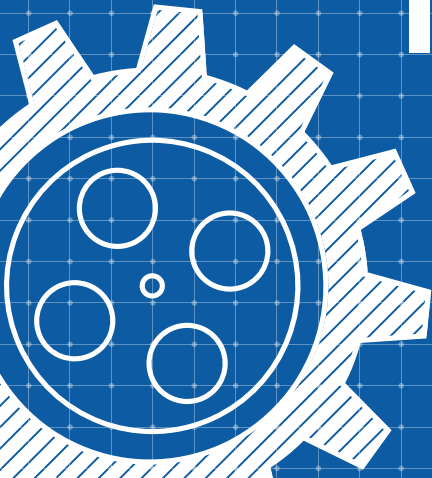


- Front, top, right and isometric views on one A size sheet

- Section view B-B shows the internal mechanism and clearances

- Includes main dimensions at scale 1:2 for reference

Design Issues and Improvements



Problems in Original Design

Wobble of top plate under load

- When we placed heavier tumblers, the platform tilted and did not stay perfectly horizontal
- Caused by a **short, thin central hub** and small contact area with the shaft

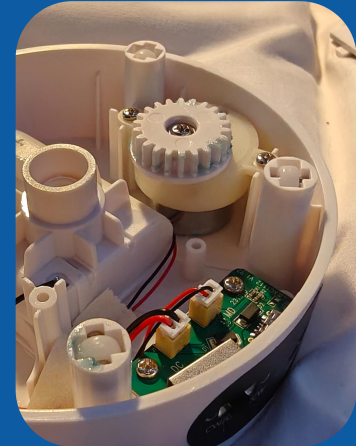


Gear teeth too small and shallow

- Under higher load the driven gear **slipped / skipped** on the motor gear
- This limited the amount of weight the stand could rotate and caused uneven motion

Limited load capacity overall

- The combination of wobble and slipping made the stand feel “cheap” and unreliable for bigger cups



Our Improvements

Improved gear for higher load

- **Enlarged tooth size and face width** of the driven gear
- Slightly increased the overall gear diameter to keep good engagement with the motor gear
- Result: **better torque transfer** and less risk of teeth slipping under load



Before

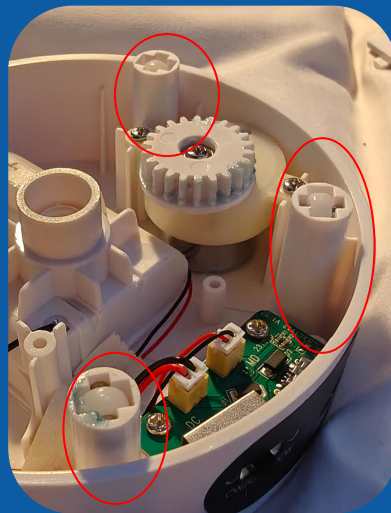


After

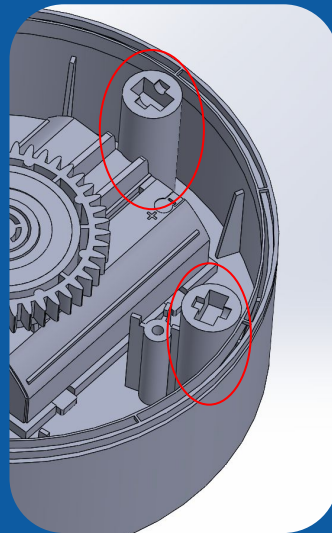
Our Improvements

Stronger support for the platform

- Increased **hub height and thickness** under the top plate
- Added **larger contact area** around the shaft and small fillets at the base
- Result: **stiffer connection**, noticeably less wobble with heavier cups



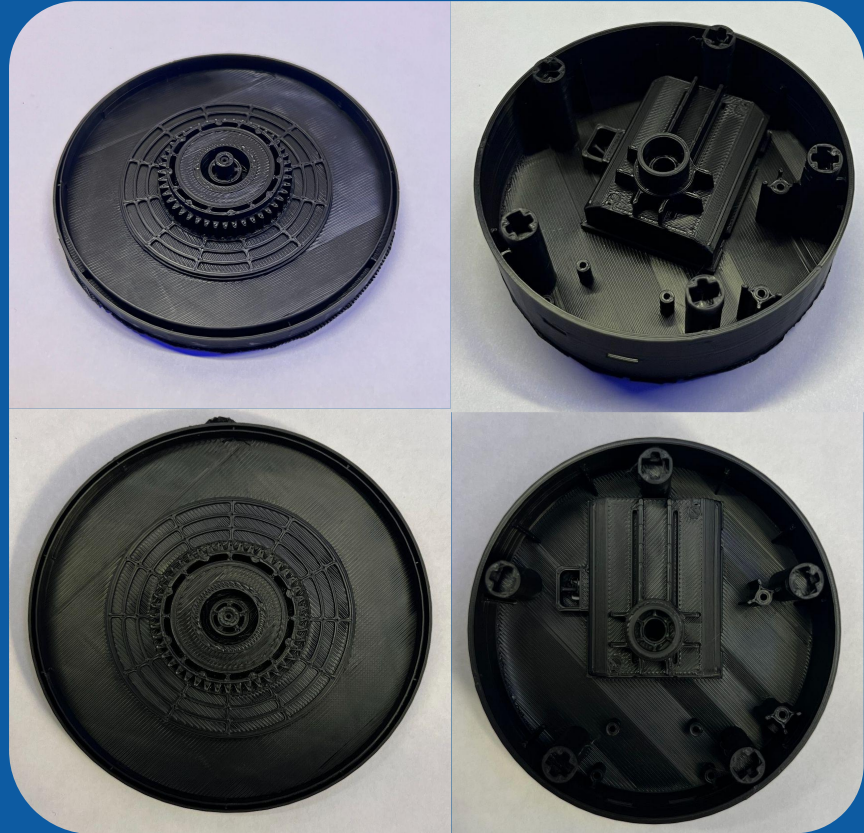
Before
11mm



After
13mm

3D Printed Parts and Testing

- Redesigned components printed in the Makerspace
- Test fit showed proper alignment with original motor and housing
- New hub reduced visible wobble when rotating a heavier tumbler
- Larger gear teeth engaged more consistently with the motor gear





3D Printed Parts & Performance

Original design (Before)

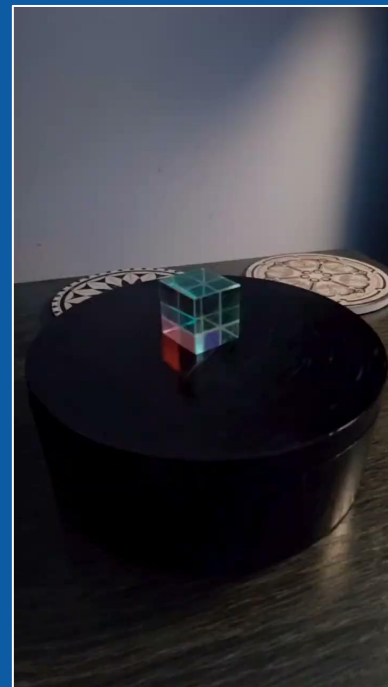
- Lower rotation speed
- Noticeable wobble under heavier loads
- Occasional gear slipping due to small gear teeth

Redesigned & 3D printed (After)

- Increased rotation speed
- Smoother and more stable rotation
- Improved load capacity with no gear slipping

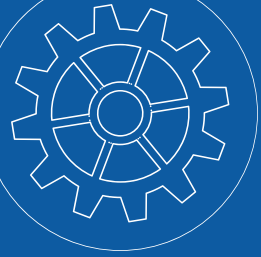


Before



After

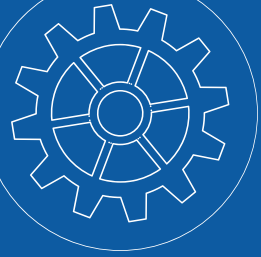




Conclusion

- Completed full reverse engineering of a rotating display stand
- Went from imagination view to 2D sketches, SolidWorks models, and assembly drawing
- Identified real mechanical issues in wobble, gear slip, and load capacity
- Proposed and implemented redesigns with 3D printed parts and test fits





Future Work

- Test with a range of cup weights and measure deflection and speed
- Explore alternative materials or thicker bodies for even more stiffness
- Analyze cost impact of changes for possible real production
- Implement a variable speed feature





Thank you & Questions?

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